Athens University of Economics and Business
Department of Statistics

Stochastic Models in Finance

Variance Reduction Techniques for Monte-Carlo Pricing:

Theory, Computation and Applications

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1 Introduction

The pricing of financial derivatives fundamentally reduces to a problem of numerical integration. Under the fundamental theorem of asset pricing, assuming an arbitrage-free market, the fair value of a derivative is the discounted expected value of its future payoff evaluated under a risk-neutral probability measure $\mathbb{Q}$. Let T represent the maturity of the contract, r the constant risk-free interest rate, and $f(S_T)$ is the payoff function which is dependent on the terminal state of the underlying asset S_T. Let C_0 represent the present value which is given by:

$$C_0 = e^{-rT}\mathbb{E}^{\mathbb{Q}}[f(S_T)] \tag{1}$$

For standard European options under the geometric Brownian motion assumption, this expectation yields closed-form analytical solutions, most notably the Black-Scholes formula. However, for high-dimensional portfolios, the expectation lacks an analytical form, necessitating numerical approximation.

The standard Monte Carlo method approaches this by simulating independent and identically distributed (i.i.d.) sample paths of the underlying asset. By defining a sequence of i.i.d. random variables $X_i = e^{-rT} f(S_T^{(i)})$ representing the discounted payoff of the i-th simulated path, we construct the Monte Carlo estimator $\hat{C}_N$ as follows:

$$\hat{C}_N = \frac{1}{N}\sum_{i=1}^{N} X_i \tag{2}$$

The theoretical justification for this approach is the Law of Large Numbers, which guarantees that as the number of simulations N approaches infinity, the sample mean converges almost surely to the true expected value: $\hat{C}_N \to C_0$.

While the Law of Large Numbers guarantees asymptotic convergence, it does not quantify the rate of that specific convergence. This error distribution is dictated by the Central Limit Theorem (CLT). Let $\sigma^2 = \mathrm{Var}(X_i)$ denote the variance of the discounted payoff. As $N \to \infty$, the standardized error converges in distribution to a standard normal variable:

$$\frac{\sqrt{N}}{\sigma}(\hat{C}_N - C_0) \xrightarrow{d} \mathcal{N}(0,1) \tag{3}$$

Consequently, the Root Mean Square Error (RMSE) of the estimator is given by:

$$\mathrm{RMSE}(\hat{C}_N) = \tfrac{\sigma}{\sqrt{N}}.$$

This formula reveals the fundamental weakness of standard Monte Carlo pricing: its extremely slow convergence. Because the error shrinks at a rate proportional to $1/\sqrt{N}$, improving accuracy is computationally expensive. Specifically, to cut the pricing error in half, one must quadruple the number of simulations N.

In industrial applications, where large portfolios of complex derivatives must be priced rapidly to assess risk exposure, scaling N exponentially introduces an unacceptable computational bottleneck. Since artificially restricting the sample size compromises pricing accuracy, the only mathematically viable solution to improve efficiency without sacrificing precision is to target the numerator of the RMSE function. We must implement variance reduction techniques to explicitly minimize σ^2.

The rest of the project is organized as follows. Section 2 covers the mathematical theory behind Antithetic and Control Variates. Section 3 explains how to transition from continuous math to a discrete computer simulation using the Euler-Maruyama scheme. Section 4 introduces our specific application, setting up the coding logic for a Down-and-Out Barrier Call. Section 5 analyzes the actual R script results, and Section 6 provides a brief conclusion.

2 Theoretical Methodology

The fundamental flaw of the naive Monte Carlo estimator, as established in the Introduction, is its convergence rate. To neutralize this computational bottleneck, we must implement Variance Reduction Techniques. The mathematical objective is to construct a modified estimator that retains the unbiased nature of the standard estimator while strictly minimizing the variance σ^2. This specific section formally details the probability framework and mathematical proofs for two standard techniques: Antithetic Variates and Control Variates.

2.1 Antithetic Variates

The method of antithetic variates is heavily utilized in financial mathematics due to its computational elegance. More specifically, this method exploits the inherent symmetry of the underlying probability distributions without requiring the evaluation of secondary financial instruments.

Mathematical Definition

In the context of asset pricing, the underlying stochastic processes are driven by standard Brownian motion, denoted as W_t. By the symmetry property of the Gaussian distribution, if W_t is a standard Brownian motion under the probability measure $\mathbb{Q}$, then its reflection, $-W_t$, is also a standard Brownian motion under $\mathbb{Q}$.

Let $f(W)$ be the discounted payoff function of a derivative evaluated along a simulated path. Instead of generating N completely independent sample paths, the antithetic method generates $N/2$ independent paths and pairs each with its exact opposite.

We define the standard path payoff as $X_i = f(W^{(i)})$ and the antithetic path payoff as $\tilde{X}_i = f(-W^{(i)})$. Because both W and $-W$ share the same distribution, the following condition will be true:

$$\mathbb{E}[X_i] = \mathbb{E}[\tilde{X}_i] = C_0 \tag{4}$$

We construct the antithetic estimator, $\hat{C}_{AV}$, by averaging these paired observations:

$$\hat{C}_{AV} = \frac{1}{N/2}\sum_{i=1}^{N/2} Y_i \quad \text{where} \quad Y_i = \frac{X_i + \tilde{X}_i}{2} \tag{5}$$

By the linearity of expectation, $\hat{C}_{AV}$ remains an unbiased estimator of C_0:

$$\mathbb{E}[\hat{C}_{AV}] = \frac{1}{2}(\mathbb{E}[X_i] + \mathbb{E}[\tilde{X}_i]) = C_0 \tag{6}$$

Variance Reduction

To demonstrate the efficiency of this method, we must evaluate the variance of the antithetic estimator and compare it against the variance of the standard Monte Carlo estimator, denoted as $\text{Var}(\hat{C}_{MC}) = \frac{\sigma^2}{N}$.

The variance of a single antithetic pair Y_i is given by:

$$\text{Var}(Y_i) = \text{Var}\left(\frac{X_i + \tilde{X}_i}{2}\right) = \frac{1}{4}\text{Var}(X_i) + \frac{1}{4}\text{Var}(\tilde{X}_i) + \frac{1}{2}\text{Cov}(X_i, \tilde{X}_i) \tag{7}$$

Since X_i and $\tilde{X}_i$ are identically distributed, $\text{Var}(X_i) = \text{Var}(\tilde{X}_i) = \sigma^2$. Substituting this into the equation simplifies the variance as follows:

$$\text{Var}(Y_i) = \frac{1}{2}\sigma^2 + \frac{1}{2}\text{Cov}(X_i, \tilde{X}_i) \tag{8}$$

The total variance of the antithetic estimator over $N/2$ pairs is therefore:

$$\text{Var}(\hat{C}_{AV}) = \frac{\text{Var}(Y_i)}{N/2} = \frac{\sigma^2 + \text{Cov}(X_i, \tilde{X}_i)}{N} \tag{9}$$

Comparing this result to $\text{Var}(\hat{C}_{MC})$, we are able to recognize the fundamental condition for the success of the antithetic variates method: variance is strictly reduced if and only if $\text{Cov}(X_i, \tilde{X}_i) < 0$.

Application to Monotonic Payoffs

Moving on to financial applications, the necessary condition of negative covariance is frequently guaranteed by the structure of the derivative's payoff function. If the payoff function $f(W)$ is monotonic with respect to the underlying Brownian motion, the variables X_i and $\tilde{X}_i$ are guaranteed to be negatively correlated. Standard European options have monotonic payoffs, meaning their value strictly moves in one direction relative to the asset price. Consequently, a simulated path that generates a high payoff is naturally balanced by its antithetic path generating a low or zero payoff. This inverse relationship stabilizes the sample mean and manages to reduce the variance σ^2.

2.2 Control Variates

While antithetic variates manipulate the internal symmetry of the simulation, the control variates method reduces variance by introducing external and highly correlated information. Specifically, it uses the mathematically proven price of a simple option to correct the simulated price of a more complex option.

The Linear Control Estimator

For the next step, let X denote the unknown and discounted payoff of the complex option we would like to price, where $\mathbb{E}[X] = C_0$. Furthermore, we introduce the control variate Y, which shares similar market dynamics and properties with X. The critical requirement is that the closed-form expected value of the control $\mathbb{E}[Y]$ is known (Black-Scholes formula).

We define the controlled estimator X_c as:

$$X_c = X - c(Y - \mathbb{E}[Y]) \tag{10}$$

where $c \in \mathbb{R}$ is a deterministic coefficient.

Because the expected value of the adjustment term is zero ($\mathbb{E}[Y - \mathbb{E}[Y]] = 0$), the controlled estimator is unbiased for any value of c:

$$\mathbb{E}[X_c] = \mathbb{E}[X] - c(\mathbb{E}[Y] - \mathbb{E}[Y]) = C_0 \tag{11}$$

Optimization of the Control Coefficient

The objective now is to select the specific coefficient c^* that manages to minimize the variance of X_c. Based on that, we expand the variance of the controlled estimator as follows:

$$\mathrm{Var}(X_c) = \mathrm{Var}(X - cY) = \mathrm{Var}(X) - 2c\mathrm{Cov}(X, Y) + c^2\mathrm{Var}(Y) \tag{12}$$

The graph of this equation is a parabola. To find the global minimum, we differentiate the variance function with respect to c and set the derivative equal to zero:

$$\frac{\partial}{\partial c}\mathrm{Var}(X_c) = -2\mathrm{Cov}(X, Y) + 2c\mathrm{Var}(Y) = 0 \tag{13}$$

Solving this for c yields the optimal control coefficient, c^*:

$$c^* = \frac{\mathrm{Cov}(X, Y)}{\mathrm{Var}(Y)} \tag{14}$$

Maximum Variance Reduction

By substituting the result above for the optimal coefficient c^*, back into the original variance equation, we can quantify the exact magnitude of the variance reduction.

$$\mathrm{Var}(X_{c^*}) = \mathrm{Var}(X) - 2\left(\frac{\mathrm{Cov}(X,Y)}{\mathrm{Var}(Y)}\right)\mathrm{Cov}(X,Y) + \left(\frac{\mathrm{Cov}(X,Y)}{\mathrm{Var}(Y)}\right)^2 \mathrm{Var}(Y) \tag{15}$$

$$\mathrm{Var}(X_{c^*}) = \mathrm{Var}(X) - \frac{\mathrm{Cov}(X,Y)^2}{\mathrm{Var}(Y)} \tag{16}$$

We can express this reduction in terms of the Pearson correlation coefficient, defined as $\rho_{X,Y} = \frac{\mathrm{Cov}(X,Y)}{\sqrt{\mathrm{Var}(X)\mathrm{Var}(Y)}}$. By factoring out $\mathrm{Var}(X)$, the final minimized variance is the following:

$$\mathrm{Var}(X_{c^*}) = \mathrm{Var}(X)(1 - \rho_{X,Y}^2) \tag{17}$$

On this result, the standard variance is multiplied by a reduction factor of $(1 - \rho_{X,Y}^2)$. Therefore, the success of the method depends entirely on identifying a control variate Y that exhibits a high absolute correlation with the target derivative X.

Regarding the practical computational implementations, the optimal coefficient c^* is rarely known, as it depends on the covariance between the exotic and standard options. Consequently, c^* must be estimated directly from the simulated sample paths, introducing a negligible bias that vanishes as $N \to \infty$.

3 Numerical Methodology

While the corresponding theoretical framework for variance reduction is constructed in continuous time using stochastic differential equations (SDEs), computational implementation strictly requires discrete-time approximation. A computer cannot simulate a continuous Brownian path. It can only generate random shocks at discrete time intervals. This following section details the numerical discretization required to transition appropriately from the continuous pricing model to an executable algorithmic procedure.

3.1 Asset Price Dynamics

To manage to simulate the underlying asset S_t, we assume it follows a Geometric Brownian Motion under the risk-neutral measure $\mathbb{Q}$. The continuous-time dynamics of the asset are described by the following SDE:

$$dS_t = rS_t dt + \sigma S_t dW_t \tag{18}$$

where r is the constant risk-free rate, σ is the constant volatility of the asset and W_t is a standard Brownian motion.

By applying Itô's Lemma to $f(S_t) = \ln(S_t)$, we are able to solve this SDE in order to find the exact asset price at any future time T, given its current price at time t:

$$S_T = S_t \exp\left(\left(r - \frac{\sigma^2}{2}\right)(T - t) + \sigma(W_T - W_t)\right) \tag{19}$$

3.2 The Euler-Maruyama Discretization Scheme

In order to code the Monte Carlo simulation, we must discretize the time horizon $[0, T]$ into M equal sub-intervals. Let $\Delta t = T/M$ be the time step in the procedure. The discrete time points are denoted as $t_k = k\Delta t$, for $k = 0, 1, \ldots, M$.

Following the Markov property representation established in the course lectures, the iterative simulation of the asset price path for the j-th simulation can be generalized as:

$$S(k+1)^j = H(k+1)^j S(k)^j \quad \text{for } k = 0, \ldots, M-1 \tag{20}$$

where $S(0)^j = S_0$ is the initial spot price.

The factor $H(k+1)^j$ represents the discrete random market shock applied to the asset over the interval Δt. By applying the Euler-Maruyama scheme to the corresponding solution of the GBM, we explicitly define the shock factor as:

$$H(k+1)^j = \exp\left(\left(r - \frac{\sigma^2}{2}\right)\Delta t + \sigma\sqrt{\Delta t} Z_{k+1}^j\right) \tag{21}$$

Here, $Z_{k+1}^j \sim \mathcal{N}(0, 1)$ are independent and identically distributed standard normal random variables, which we generated algorithmically. The term $\sqrt{\Delta t} Z_{k+1}^j$ is the discrete equivalent of the Brownian increment dW_t.

3.3 Algorithmic Execution

For each simulated path j, the algorithm iterates the Euler-Maruyama step M times to construct the full trajectory of the asset. Once the terminal price $S(M)^j$ is generated, the sample payoff $F(S^j)$ is calculated.

The final standard Monte Carlo price at time $t = 0$ is computed by discounting the sample mean of the payoffs of the following form:

$$P(0) = (1+r)^{-T} \frac{1}{N} \sum_{j=1}^{N} F(S^j) \tag{22}$$

Worth mentioning is that while continuous discounting e^{-rT} is standard in the theoretical formulations, discrete compounding $(1+r)^{-T}$ is strictly utilized here to align with the specific numerical constraints of the computational framework.

To implement Variance Reduction Techniques computationally, we modify the random normal input Z (for example pairing Z with $-Z$) before calculating the market shock H. This directly reduces the variance of the final estimated price while keeping the core simulation procedure entirely unchanged.

4 Application and Coding Architecture

In order to test how well these variance reduction techniques work in practice, we apply them to a path-dependent derivative. Specifically, we will price a Down-and-Out Barrier Call Option. This is a suitable choice because it requires tracking the full simulated price path from the Euler-Maruyama scheme, making it computationally heavier and more volatile than a standard European option.

4.1 Down-and-Out Call

A Down-and-Out Call (DOC) option acts exactly like a standard European call, but with one major difference: if the asset price S_t drops below a predetermined barrier level B at any point during its life, the option instantly becomes worthless.

Let K be the strike price, T the maturity, and B the barrier level, where we assume the initial stock price starts above the barrier ($S_0 > B$). The discounted payoff function X for the DOC is defined as:

$$X = e^{-rT} \max(S_T - K, 0) \cdot \mathbb{I}(\min_{0 \le t \le T} S_t > B) \tag{23}$$

where $\mathbb{I}(\cdot)$ is the indicator function. It equals 1 if the minimum stock price of the generated path stays strictly above the barrier B, and equals 0 if the price drops below the barrier.

Because of this specific barrier condition over a discrete timeline, Monte Carlo simulation is an effective way to estimate its fair price $\mathbb{E}[X]$.

4.2 Algorithmic Architecture

The numerical simulation will be implemented in R. To ensure our code is efficient and mathematically sound, we break down the logic into four distinct computational steps.

Step 1: Standard Path Generation

Using the discrete Euler-Maruyama scheme which was described in Section 3, we first generate a matrix of standard normal random variables Z. We use these variables to build N independent price paths for the asset for M time steps. For each possible path, the code checks the minimum price reached. If this minimum is strictly greater than B, the payoff is calculated as $\max(S_T - K, 0)$. Otherwise, the payoff is recorded as 0.

Step 2: Antithetic Variates Implementation

To apply the antithetic variates method, we do not need to generate new random numbers. We simply take the original random matrix Z and multiply it by -1 to create $-Z$. As next step, we build a second set of N price paths using this mirrored matrix. The final estimator averages the payoffs from the normal paths and the mirrored paths, which manages to smooth out extreme market movements and reduces the total variance.

Step 3: Control Variates Implementation

For the control variate method, we need a simpler option Y that is highly correlated with our barrier option X, and has a known price $\mathbb{E}[Y]$. A standard European Call option with the exact same strike K and maturity T is the perfect candidate, as their payoffs are identical in the cases where the barrier threshold is not reached.

For every simulated path, the algorithm calculates both the barrier option payoff and the standard European call payoff. We compute the exact Black-Scholes price of the European call in order to guide us in the procedure. Furthermore, the code calculates the sample covariance between the two sets of payoffs to find the optimal control coefficient c^*. Finally, we adjust our barrier option estimates using the control formula derived in Section 2.

Step 4: Results

The R script will output three separate estimated prices and three separate variances: one for the naive Monte Carlo, one for the Antithetic approach, and one for the Control Variate approach. By comparing these final variances, we will demonstrate the computational efficiency gained from each technique.

5 Implementation and Results

5.1 Computational Setup

Before executing the numerical simulation, it is necessary to establish a financially realistic testing environment. The Monte Carlo procedure was programmed in R, utilizing base functions to bypass the computational inefficiencies of iterative loops and minimize the execution time.

As our initial step, we set the initial asset price to $S_0 = 100$ and the strike price of the Down-and-Out Barrier Call to $K = 100$ as well. The knock-out barrier was defined at $B = 85$. The environment where we operate assumes a constant risk-free rate of $r = 0.05$ and an asset volatility of $\sigma = 0.2$.

The time horizon of the contract is exactly one year ($T = 1.0$). Worth mentioning is that in order to mirror the discrete monitoring of real-world barrier options, the Euler-Maruyama discretization was divided into $M = 252$ steps, representing the number of trading days in a financial year. Finally, to ensure robust asymptotic convergence and leverage the Law of Large Numbers established in Section 1, the simulation size was defined to $N = 100.000$ independent sample paths.

5.2 Execution Strategy

To guarantee a fair comparison, all the three pricing methodologies must be executed to the exact same market noise.

The corresponding script first generates a base matrix of 100.000×252 standard normal random variables (Z) to construct the fundamental asset paths and compute the standard Monte Carlo baseline price. For the Antithetic Variates approach, this matrix is deterministically inverted to $-Z$, generating the mirrored paths without introducing new random variation. Regarding the Control Variates approach, the baseline simulated paths are evaluated under a standard European Call payoff structure. Then we use the Black-Scholes price to calculate the optimal coefficient c^*. By keeping the random numbers identical across the three methods, we guarantee that the drop in variance is directly caused by our techniques rather than just statistical noise.

5.3 Variance Reduction Analysis

The corresponding results of the simulation are summarized in Table 1 below.

Methodology	**Price (C_0)**	**Variance (σ^2)**	**Reduction**
Standard Monte Carlo	9.9510	216.7461	-
Antithetic Variates	10.0101	58.3868	73.06%
Control Variates	10.0457	6.4691	97.02%

Table 1: Performance of Variance Reduction Techniques

The results from Table 1 validate the theoretical proofs established in Section 2. All three methodologies successfully preserved the unbiased nature of the estimator, managing to keep the expected value tightly around 10.00.

However, their computational efficiency differs significantly. The Antithetic Variates method successfully eliminated 73.06% of the variance simply by pairing each random shock with its exact opposite, which naturally balances out unusual price movements. The Control Variates method delivered exceptional performance, achieving a 97.02% variance reduction. This massive drop in variance happens because the Barrier Call and the standard European Call are highly correlated. Since the barrier ($B = 85$) is set far below the strike ($K = 100$), their final payoffs are identical in almost every profitable scenario.

By applying these techniques, we manage to overcome the limitations of the Central Limit Theorem and also achieve precise derivative prices without forcing the computer to run millions of extra paths.

6 Conclusion

This project demonstrated that standard Monte Carlo simulations, while mathematically unbiased, are computationally inefficient for pricing path-dependent derivatives. After applying variance reduction techniques to a Down-and-Out Barrier Call option, we proved that pricing precision can be significantly improved without relying on huge computational power.

The results above speak for themselves. The Antithetic Variates method provided a highly effective baseline adjustment, cutting the variance by over 73% simply by balancing the market shocks. However, the Control Variates method was the optimal approach for this specific pricing problem. By leveraging a highly correlated standard European Call as a baseline, the algorithm successfully managed to eliminate 97% of the pricing variance.

In conclusion, simply running more simulations by scaling N is an inefficient way to reduce pricing errors. These variance reduction methods achieve the exact same accuracy with a fraction of the computational effort.

7 References

Pages, Gilles. “Numerical probability.” Universitext, Springer (2018).